

Agenda of Presentation :

“ STEERING SYSTEM IN
AUTOMOBILE ”

Definition :

The Steering is the term applied to the collection of components and linkages, etc. which will allow a vehicle to follow the desired path.

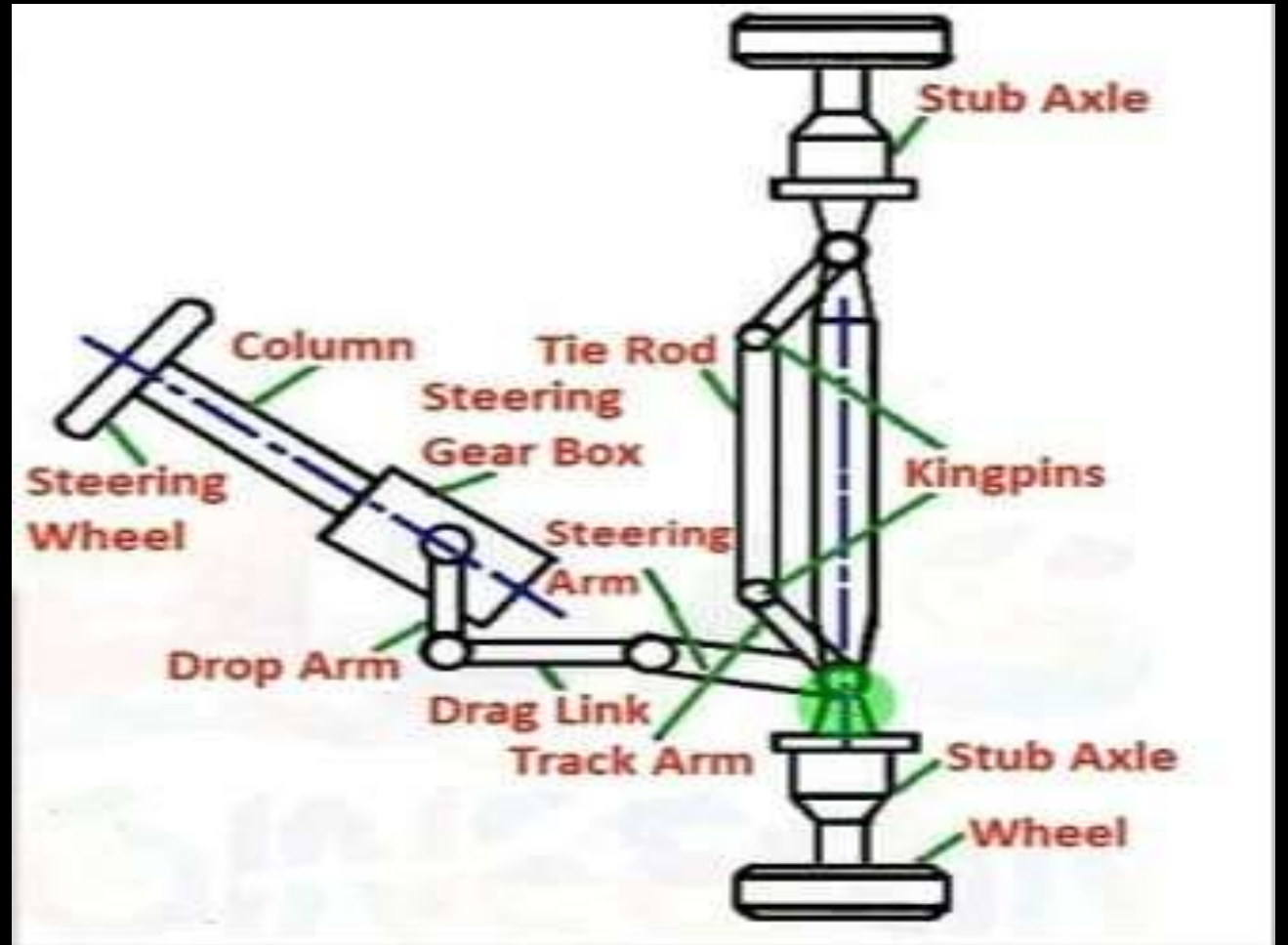
The **Steering ratio** is the ratio of number of degree of turn of steering wheel to the number of degrees the wheel turn as a result.

Functions :

1. To achieve angular motion of front wheels by rotary motion of steering wheel.
2. To provide directional stability of the vehicle when going straight ahead.
3. To provide perfect steering condition, i.e., perfect rolling motion of the road wheels at all times.
4. To facilitate straight ahead recovery after completing a turn.
5. To minimize tyre wear.

Steering linkages

- Steering Wheel
- Steering Column
- Steering Gear
- Pitman Arm
- Drag Link
- Steering Knuckle Arm
- Tie Rod
- Kingpins
- Stub Axle



Steering system types :

1. Manual Steering System
2. Power Steering System

1. Manual Steering System

In manual steering system manual force is used for steering. There are several different types of manual steering systems, which are as follows:

- i. Worm and sector
- ii. Worm and roller
- iii. Cam and lever
- iv. Worm and nut (re-circulating ball type)
- v. Rack and pinion

2. Power Steering System

The power steering system provides automatic hydraulic assistance to the turning effort applied to the manual steering system .

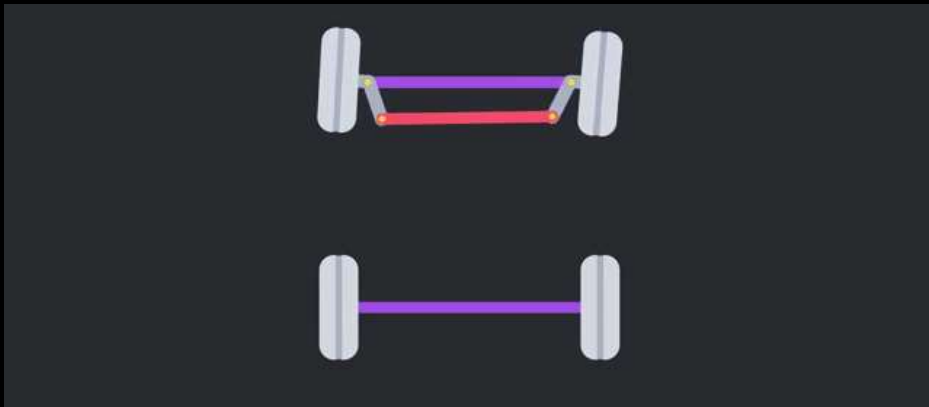
The main components of this system are.

1. Pump.
2. Rotary valve.
3. Steering mechanism

Steering Mechanism

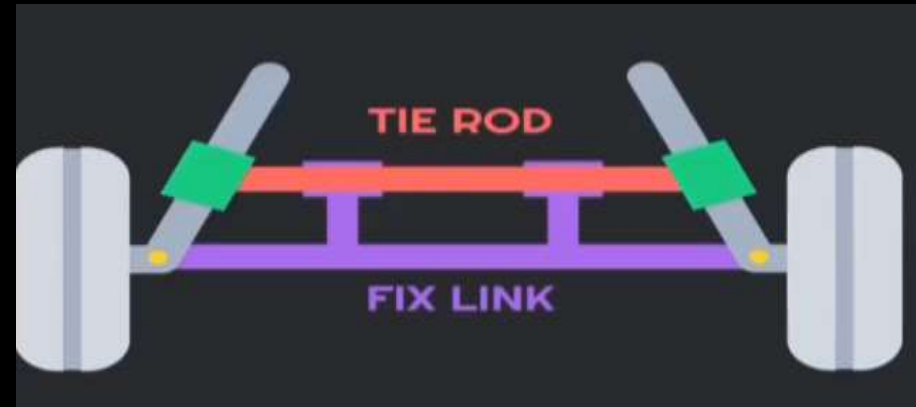
Ackermann Mechanism

The Ackermann steering mechanism is a geometric arrangement of linkages in the steering of a vehicle designed to turn the inner and outer wheels at the appropriate angles.



Davis Mechanism

Davis steering gear has two sliding pairs and two turning pairs. In this mechanism, the slotted links are attached to the front wheel axle, which turn about two pivotal points.



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THANK YOU